DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE489
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Project Title: CourtSide — Basketball Stats Tracker

Project Features:

By the end of this course, implemented features include the following:

A. Three-role authentication system (Coach / Admin / Super Admin) with SHA-256 + salt password hashing, session persistence, and disabled-account enforcement at the SQL level

B. Coach workflow—teams and players CRUD with cascade deletes, game creation with pre-initialized stat rows, live game recording with action buttons (2pt/3pt/FT made/missed), separate opponent score strip, and single-step undo

C. Game summary with computed FG%, 3P%, FT% per player; season analytics with W/L cards, points-per-game bar chart (color-coded by result), and player leaderboard; individual player profiles with shooting percentage breakdowns and per-game charts

D. PDF export for game summary, season analytics, and player profile (save / print / share via Android print preview); screenshot share for game summaries through the system share sheet

E. Admin oversight panel—system overview dashboard, coach search and drill-down (read-only), disable/enable coach accounts, raise flags on teams or games for super admin review

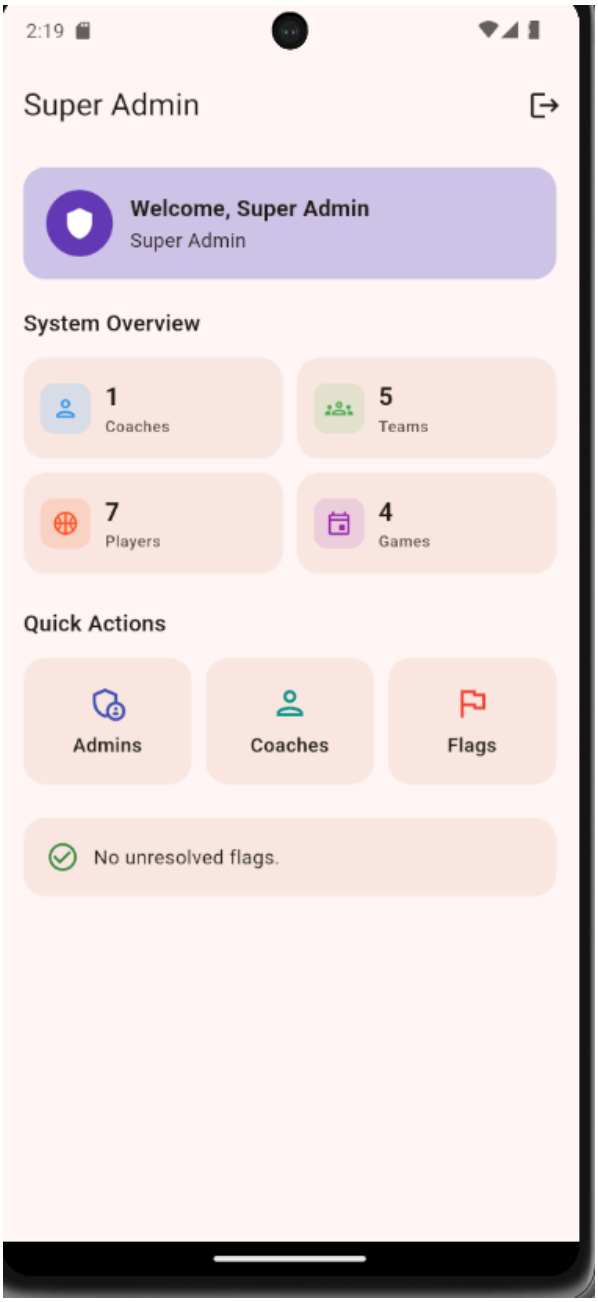
F. Super Admin panel—admin CRUD with manual or auto-generated passwords, full cascade-delete authority over coaches/teams/games, flag queue with dismiss or delete-target resolution

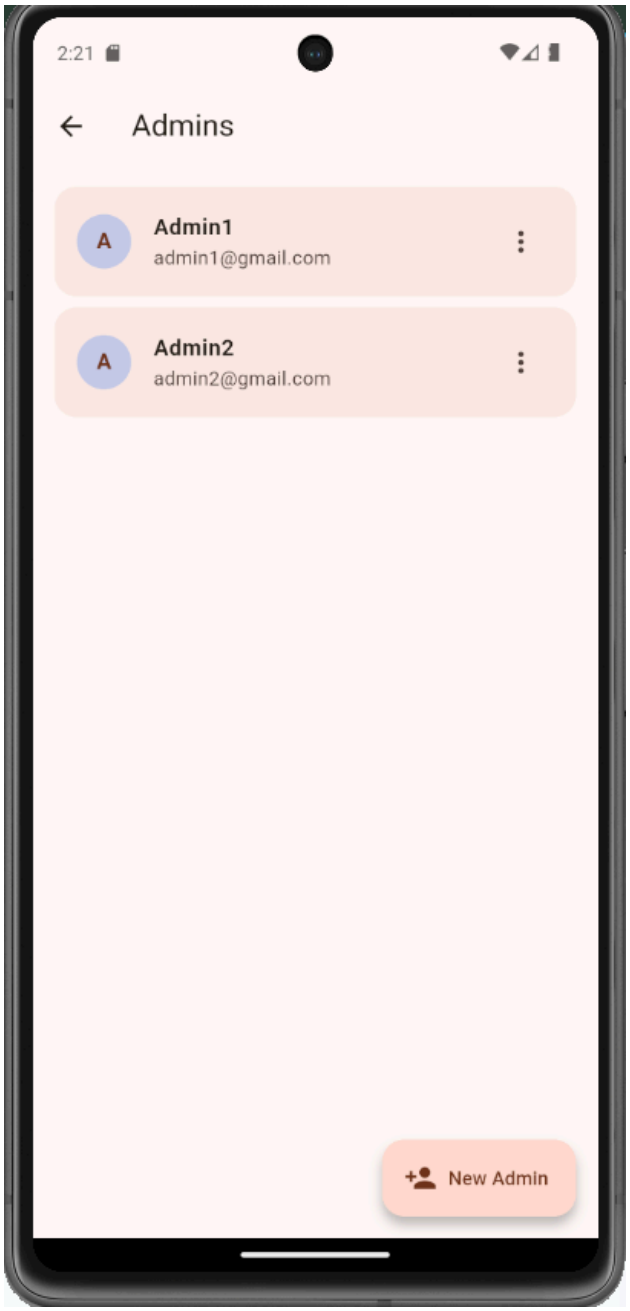
G. Reactive UI throughout—all lists and counters auto-update via streaming queries (drift. `watch()` + Riverpod StreamProvider), no manual refresh anywhere in the app

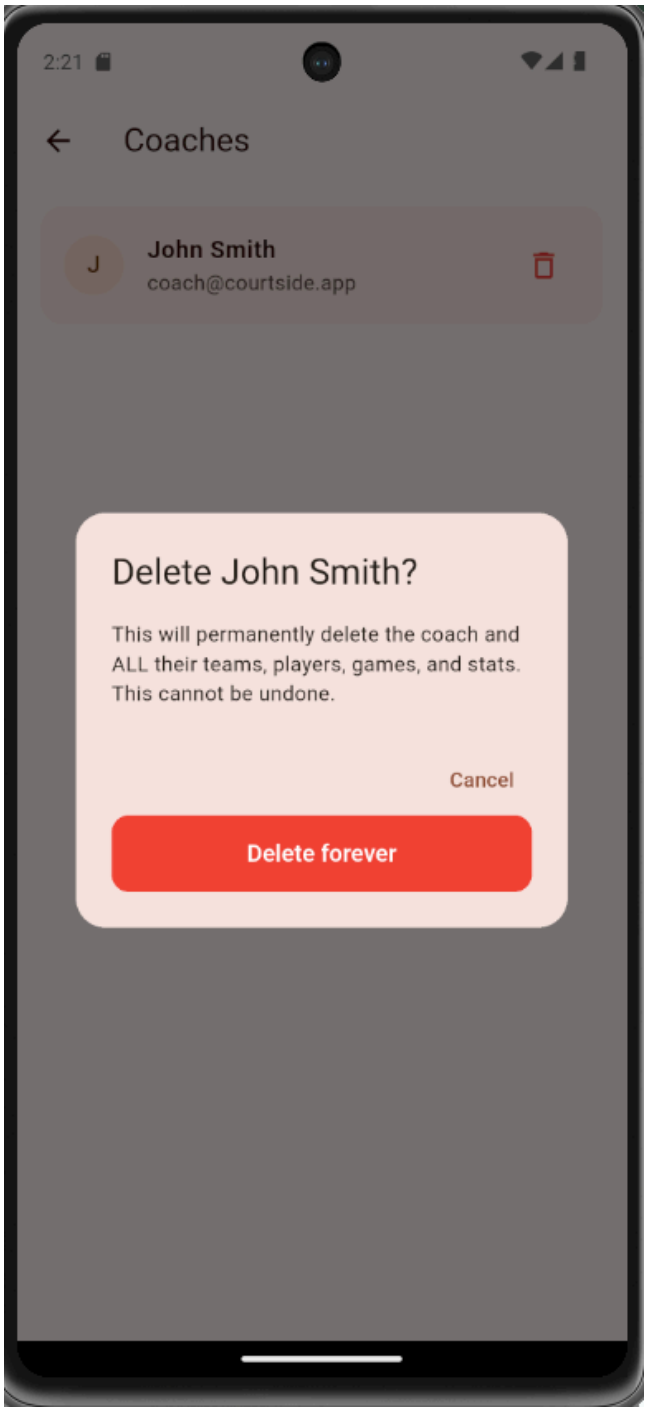
H. Demo accounts seeded on firstlaunch: Super Admin (`superadmin@courtside.app` / `SuperAdmin@2026`) and Coach (`coach@courtside.app` / `Coach@2026`)

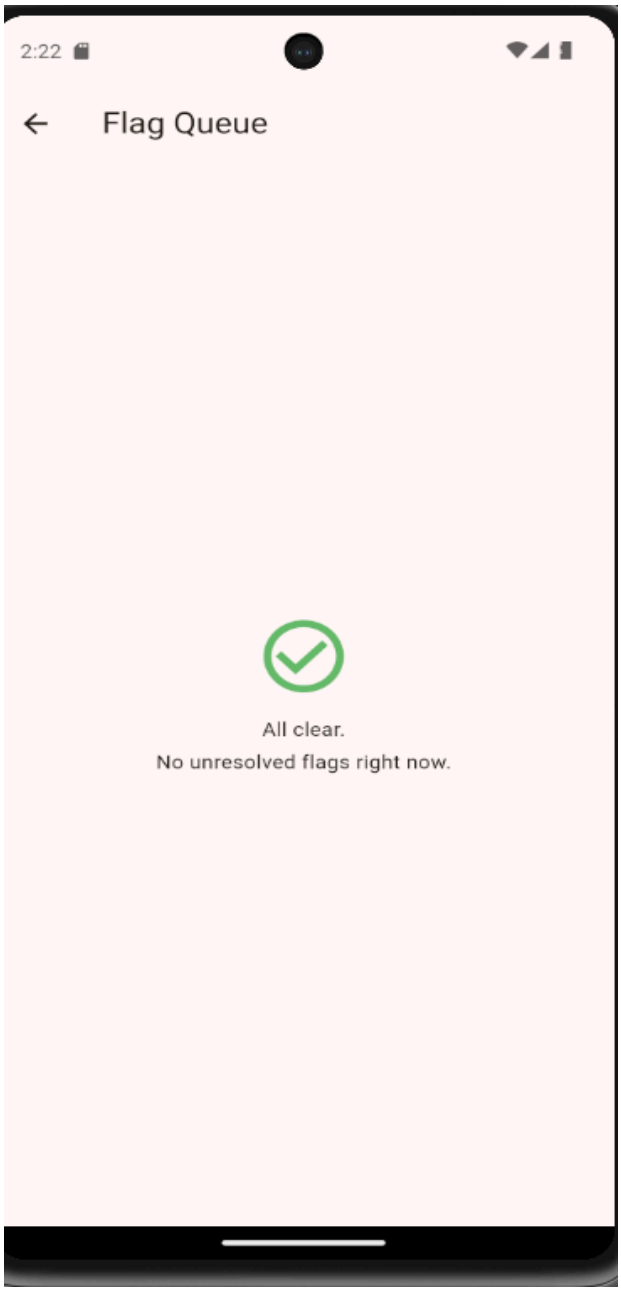
Screenshots:

Super Admin DashBoard:

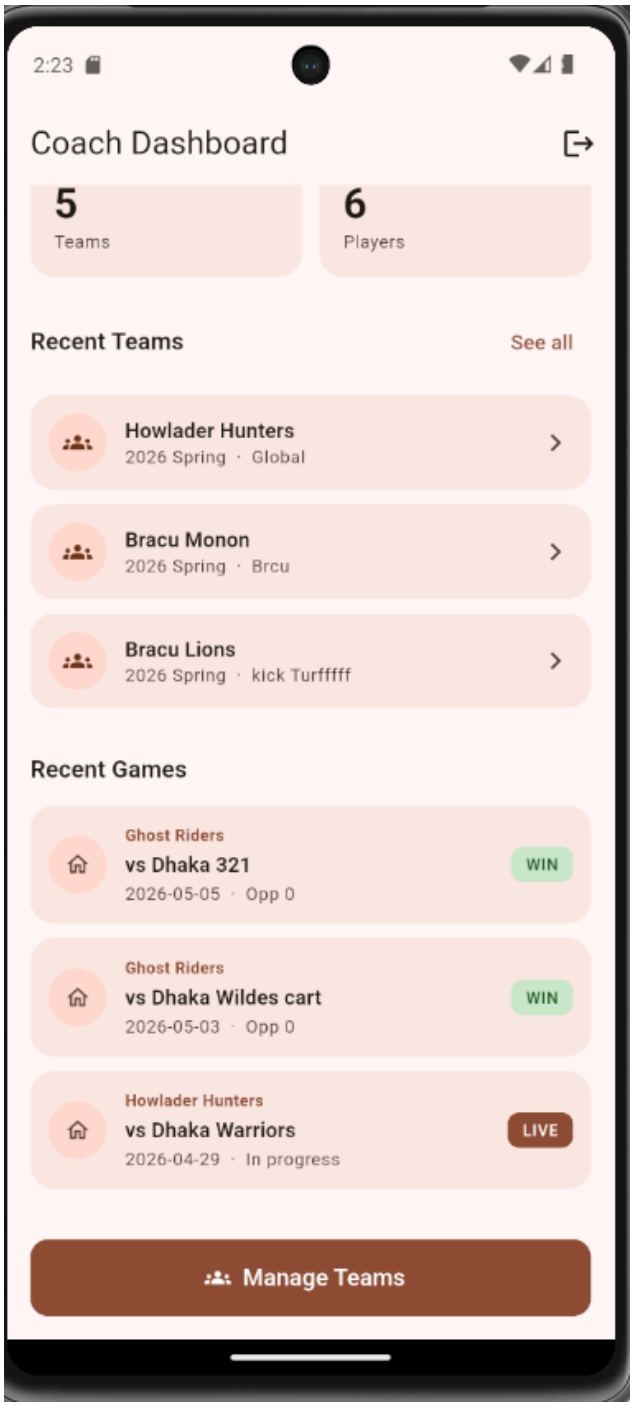


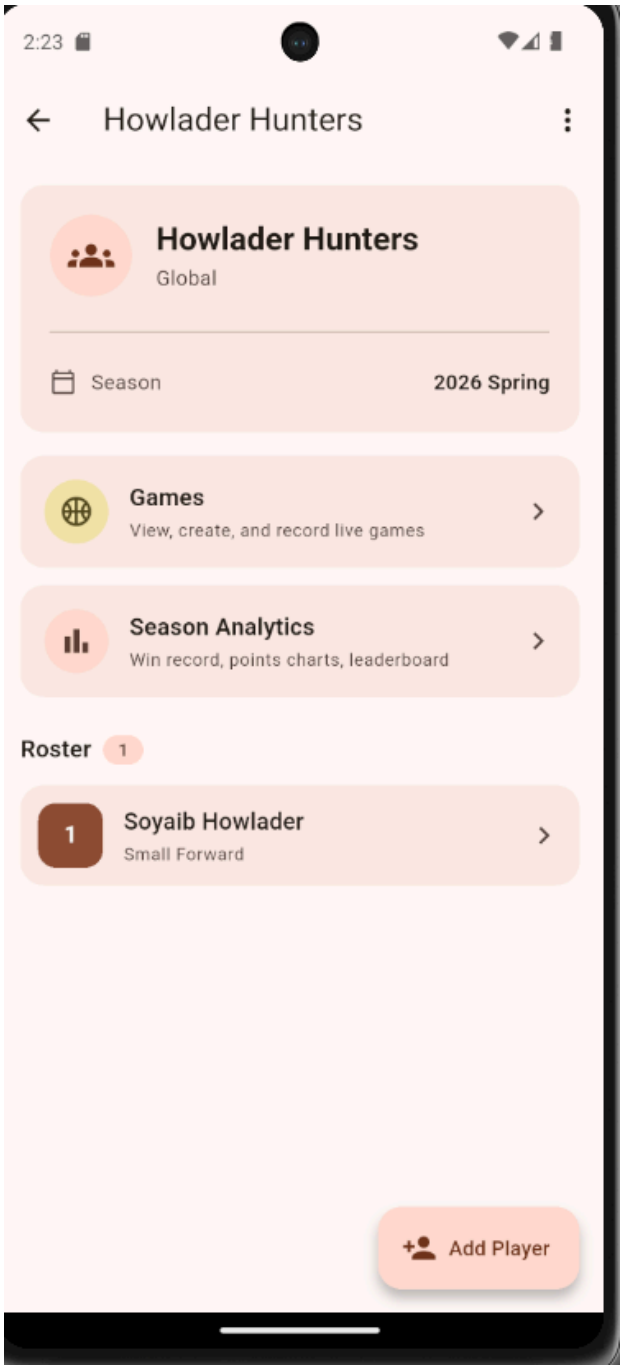

2:21
Admins
A
Admin1
admin1@gmail.com
A
Admin2
admin2@gmail.com
New Admin

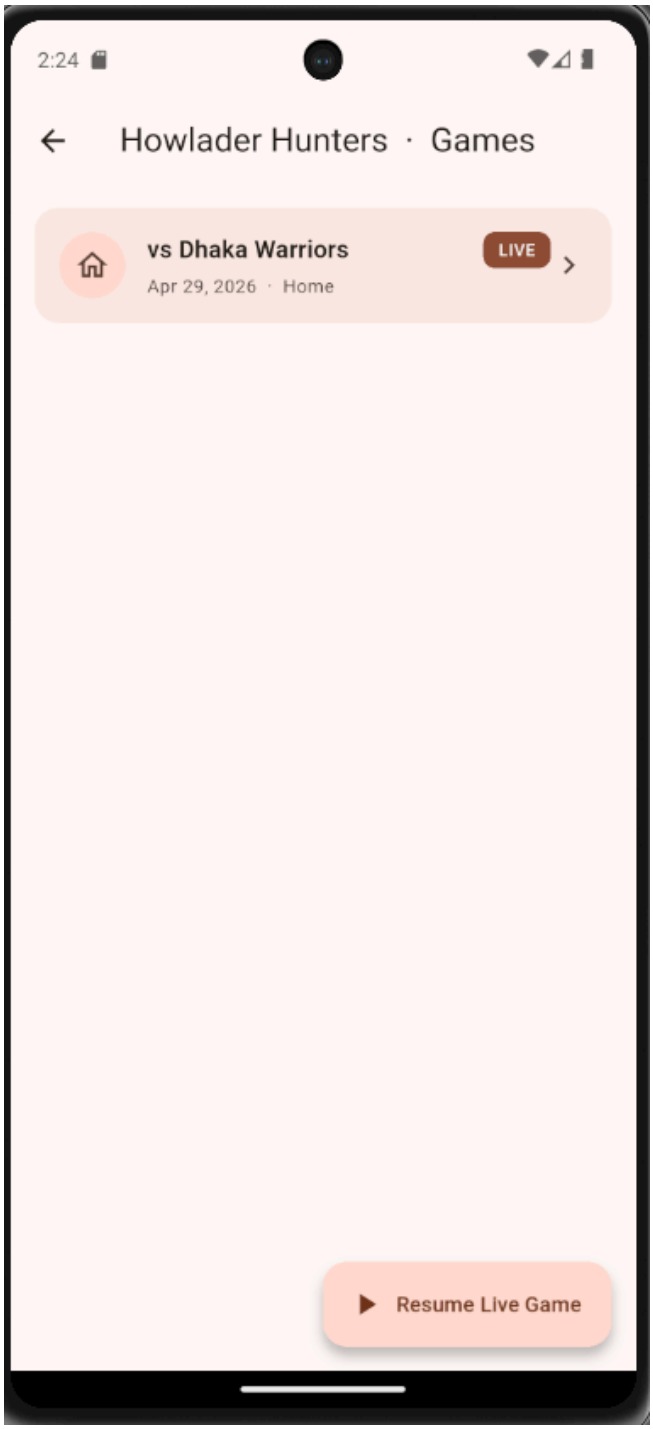

2:21
Coaches
J
John Smith
coach@courtside.app
Delete John Smith?
This will permanently delete the coach and ALL their teams, players, games, and stats. This cannot be undone.
Cancel
Delete forever

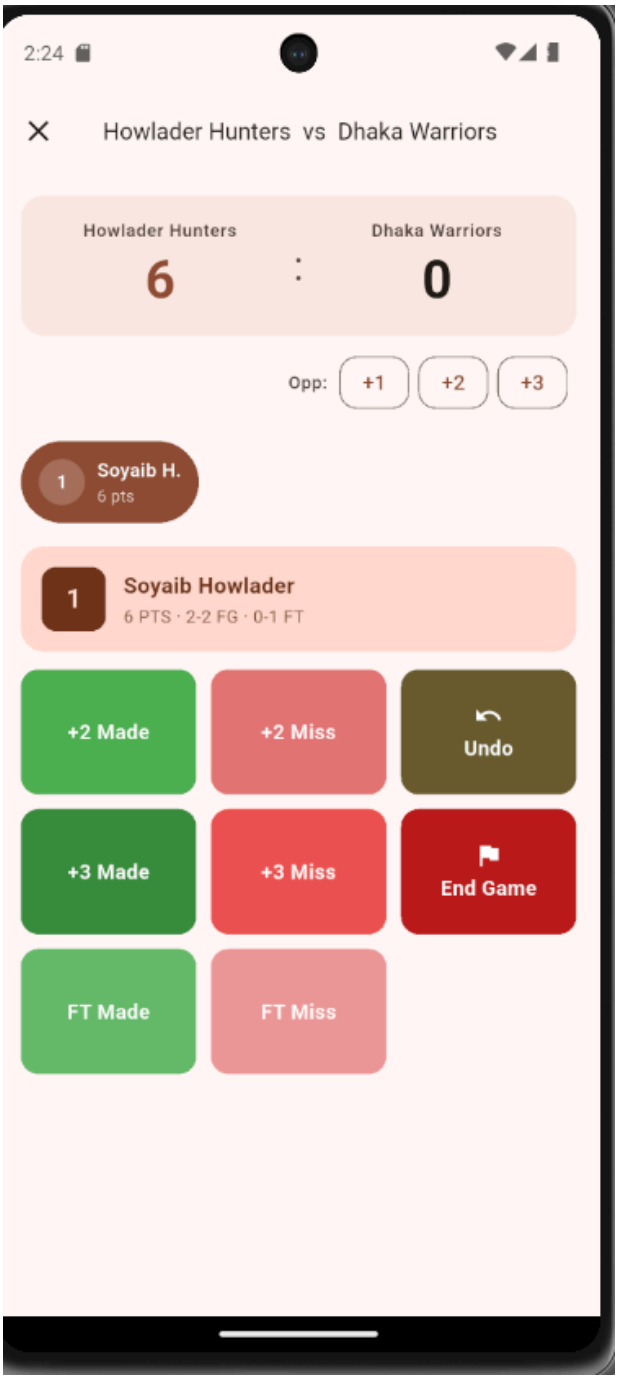

2:22
Flag Queue
All clear.
No unresolved flags right now.

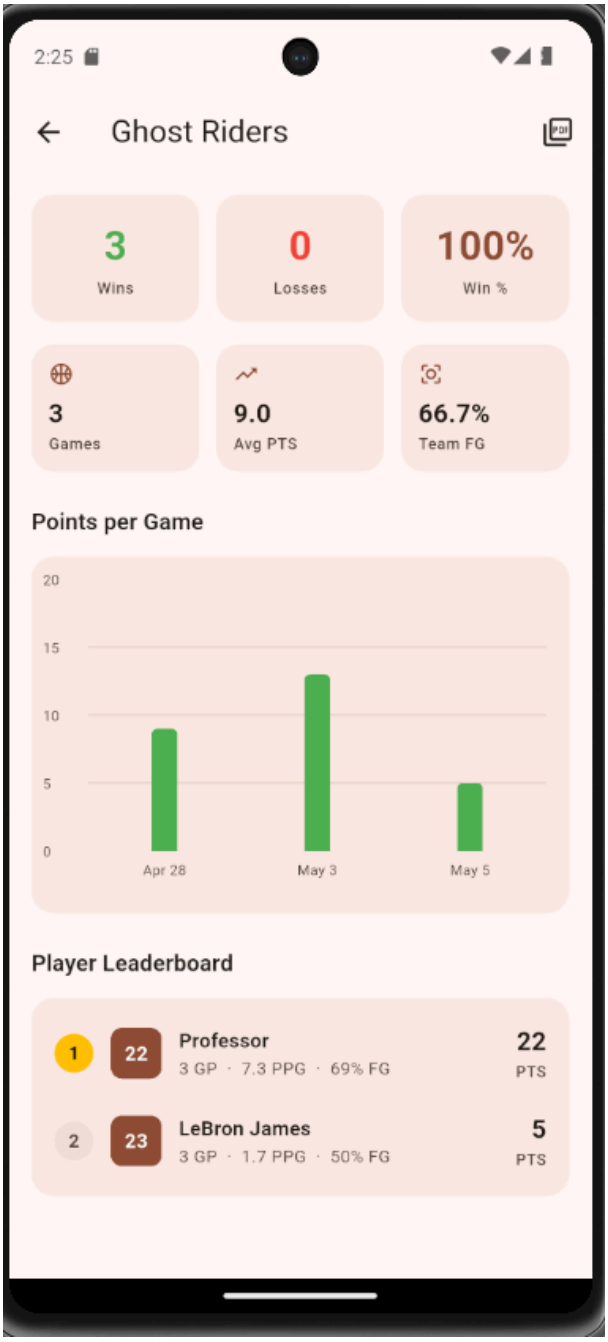
Coach Dashboard:

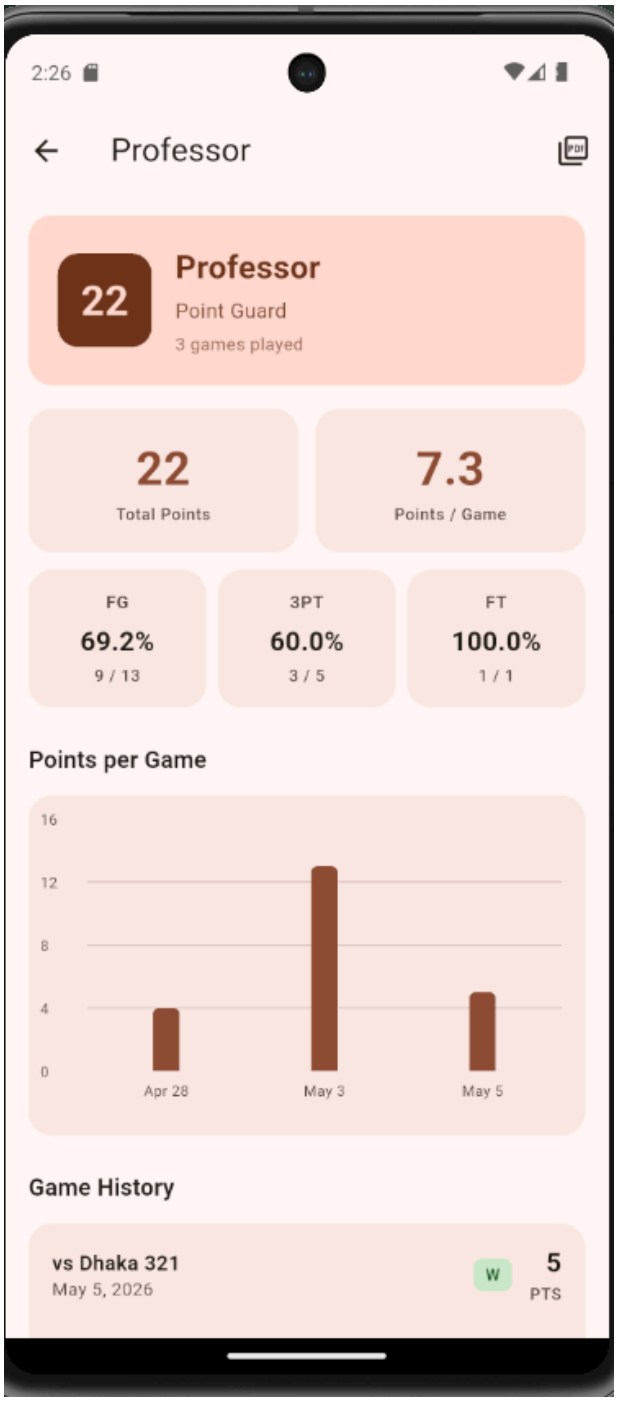


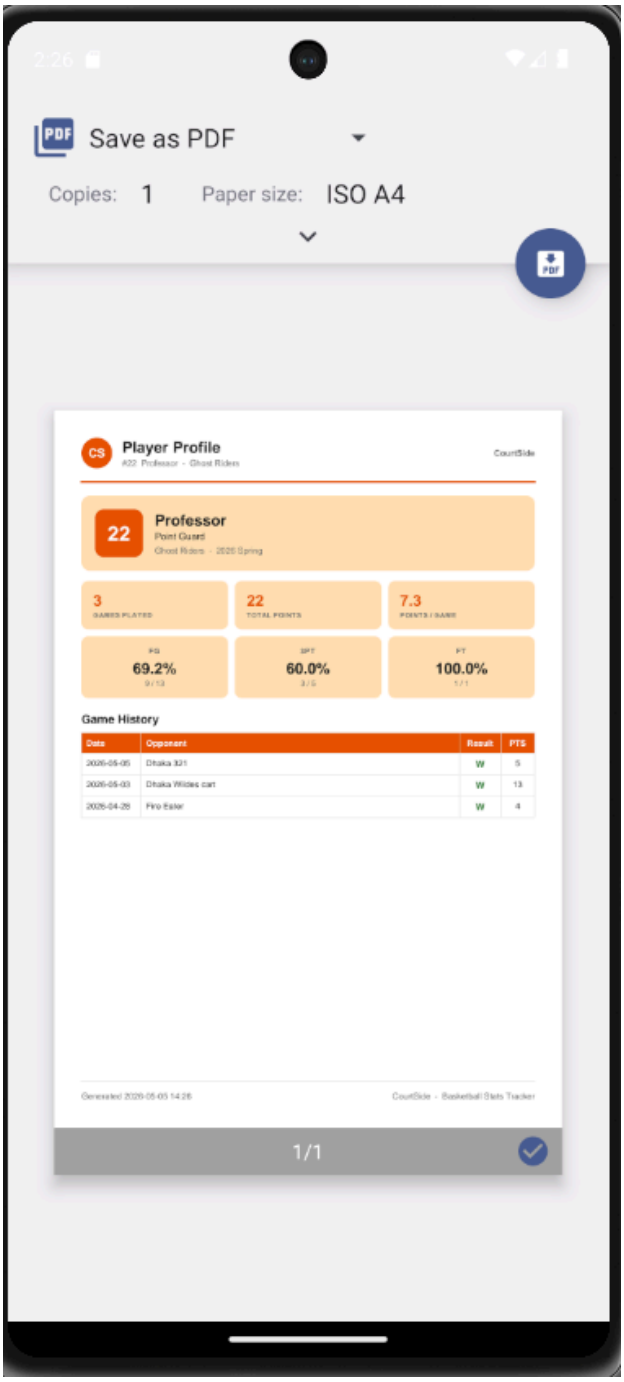

2:23
Howlader Hunters
Howlader Hunters
Global
Season
2026 Spring
Games
View, create, and record live games
Season Analytics
Win record, points charts, leaderboard
Roster
1
1
Soyaib Howlader
Small Forward
Add Player

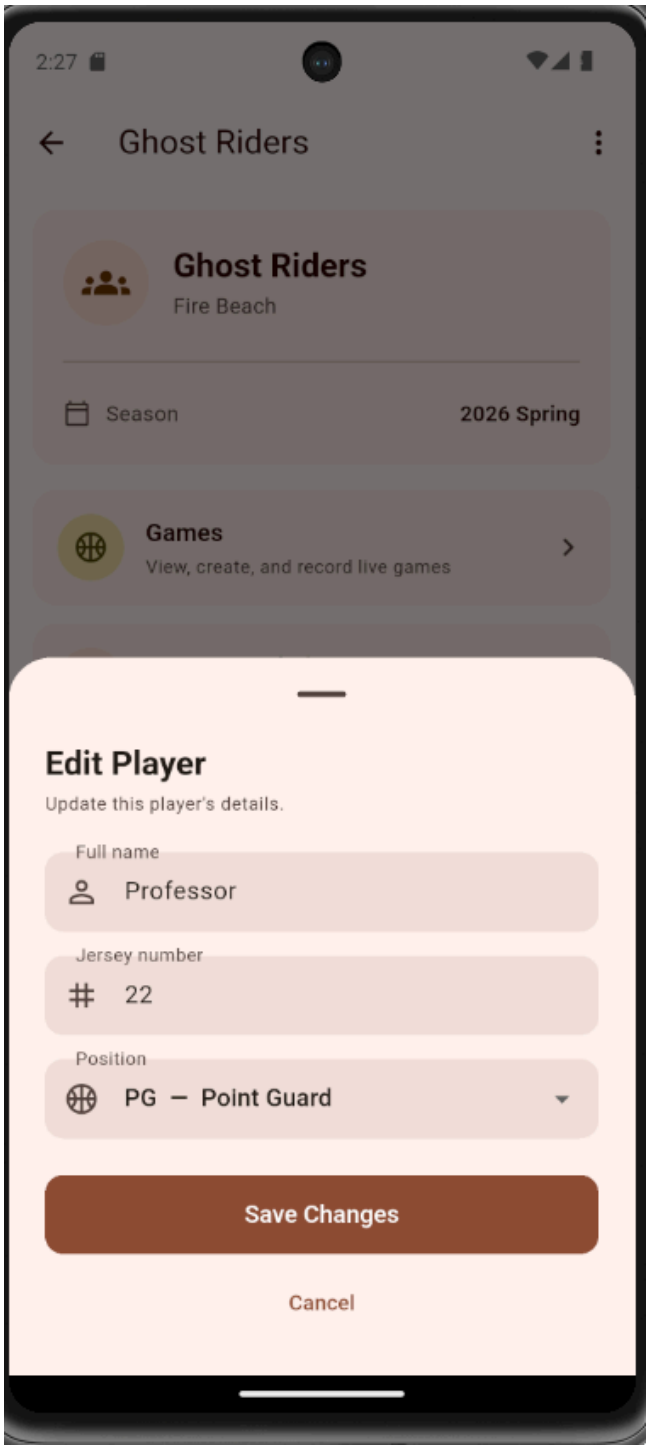

2:24
Howlader Hunters · Games
vs Dhaka Warriors
Apr 29, 2026 · Home
LIVE
Resume Live Game

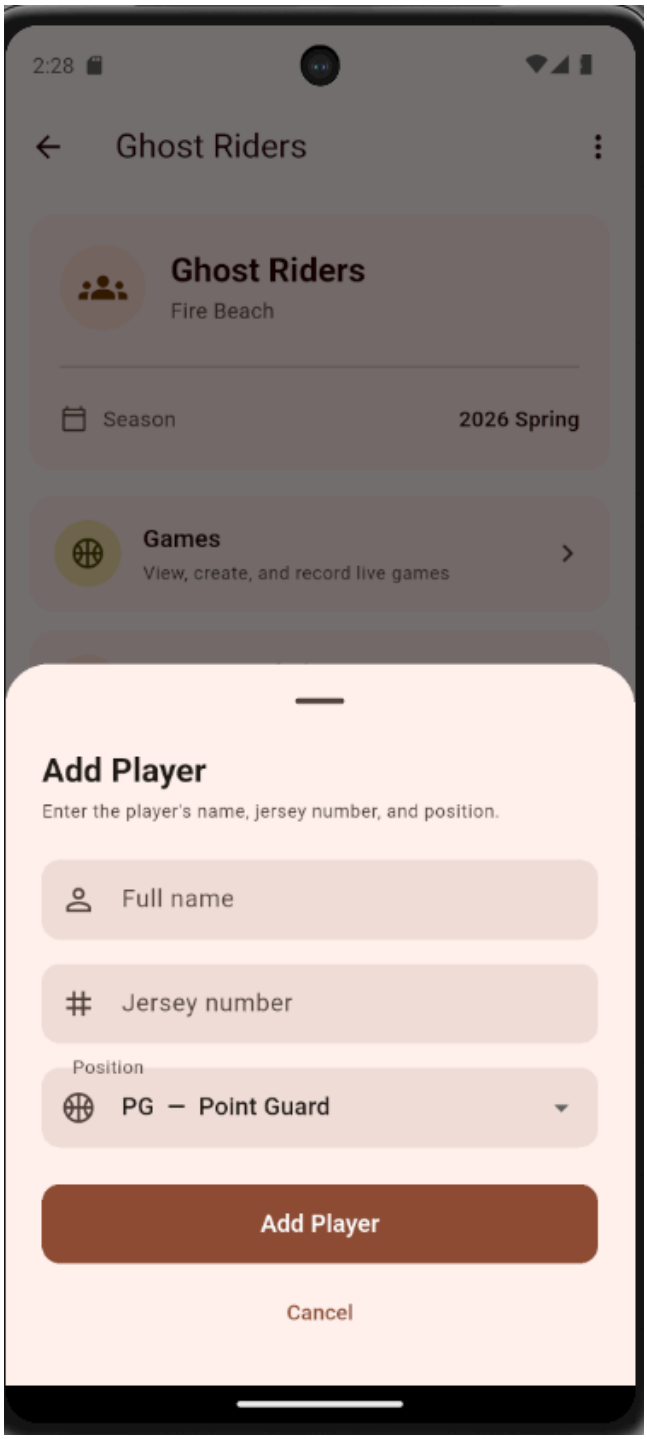

2:24
Howlader Hunters vs Dhaka Warriors
Howlader Hunters
6
:
Dhaka Warriors
0
Opp:
+1
+2
+3
1
Soyaib H.
6 pts
1
Soyaib Howlader
6 PTS · 2-2 FG · 0-1 FT
+2 Made
+2 Miss
Undo
+3 Made
+3 Miss
End Game
FT Made
FT Miss

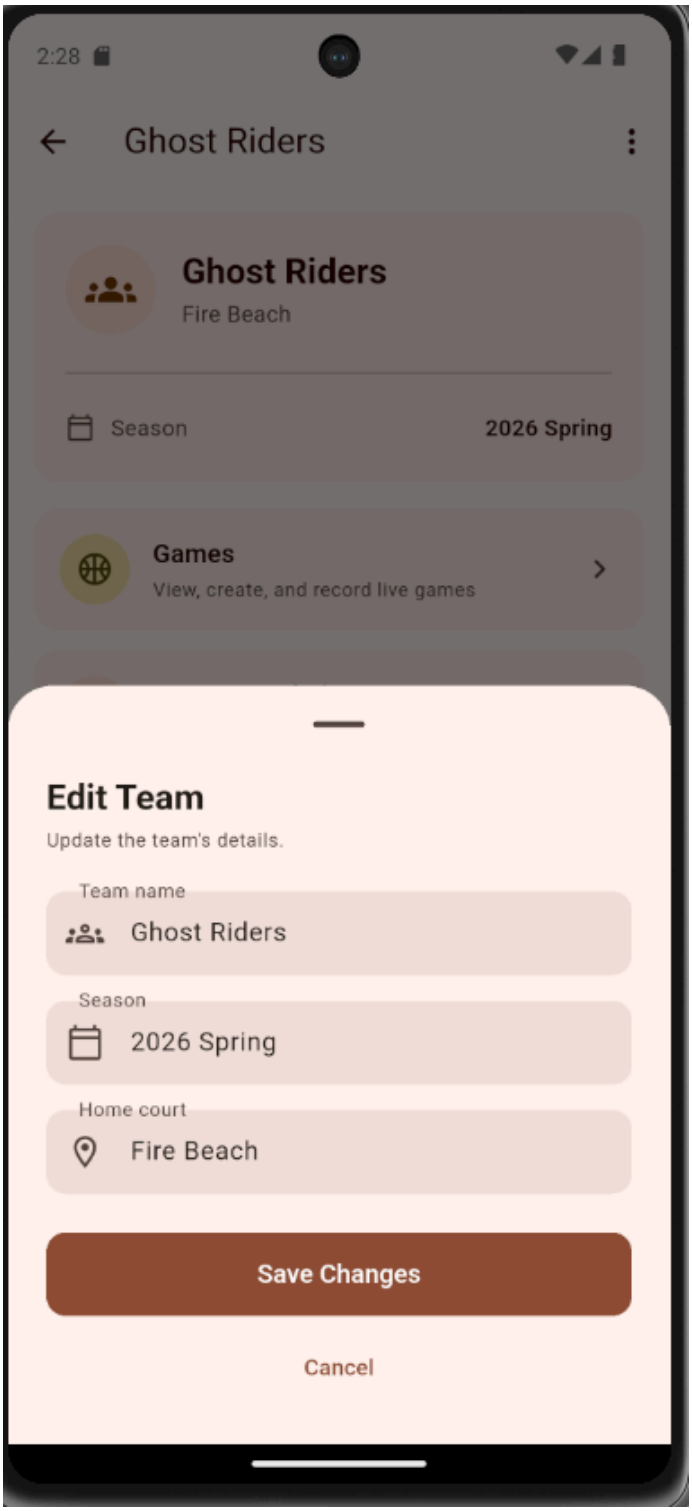

2:25
Ghost Riders
3
Wins
0
Losses
100%
Win %
3
Games
9.0
Avg PTS
66.7%
Team FG
Points per Game
20
15
10
5
0
Apr 28
May 3
May 5
Player Leaderboard
1
22
Professor
3 GP · 7.3 PPG · 69% FG
22
PTS
2
23
LeBron James
3 GP · 1.7 PPG · 50% FG
5
PTS

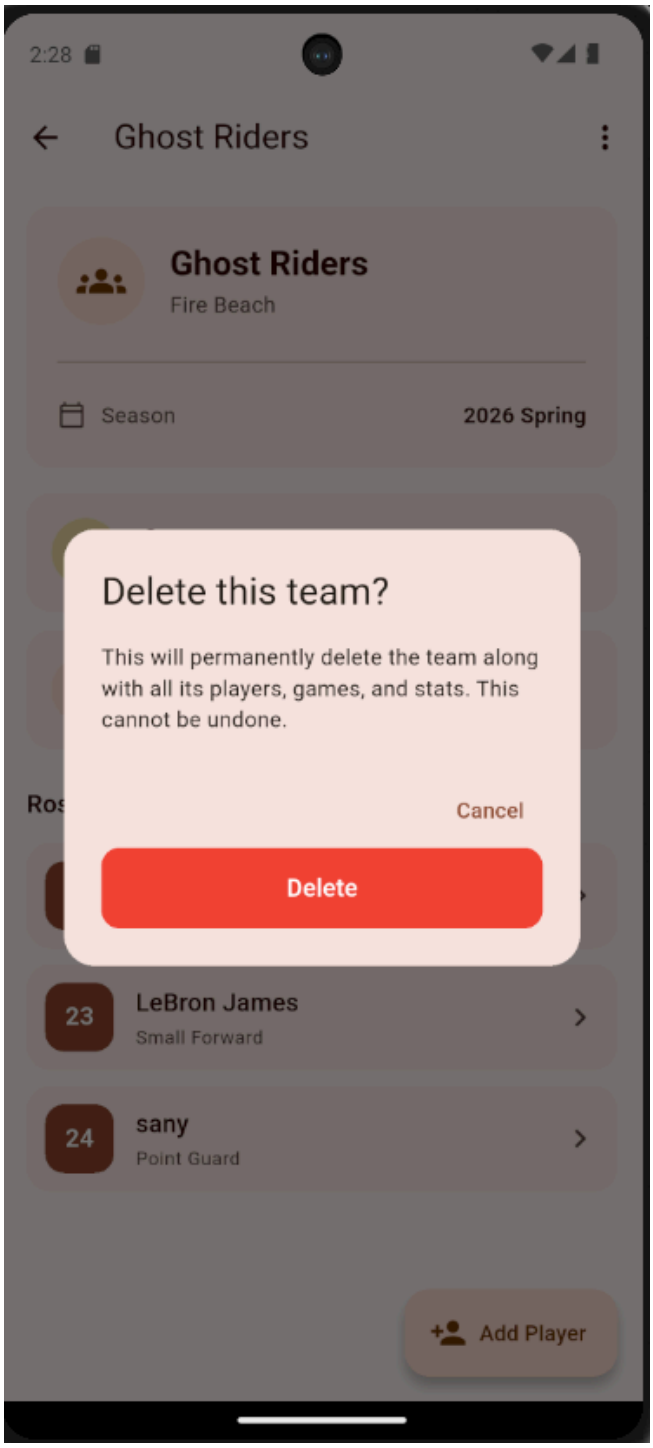

2:26
Professor
22
Professor
Point Guard
3 games played
22
Total Points
7.3
Points / Game
FG
69.2%
9 / 13
3PT
60.0%
3 / 5
FT
100.0%
1 / 1
Points per Game
16
12
8
4
0
Apr 28
May 3
May 5
Game History
vs Dhaka 321
May 5, 2026
W
5
PTS


Save as PDF
Copies: 1
Paper size: ISO A4
Player Profile
CourtSide
22
Professor
Point Guard
3
22
7.3
69.2%
60.0%
100.0%
Game History
Date
Opponent
Result
PTS
1/1

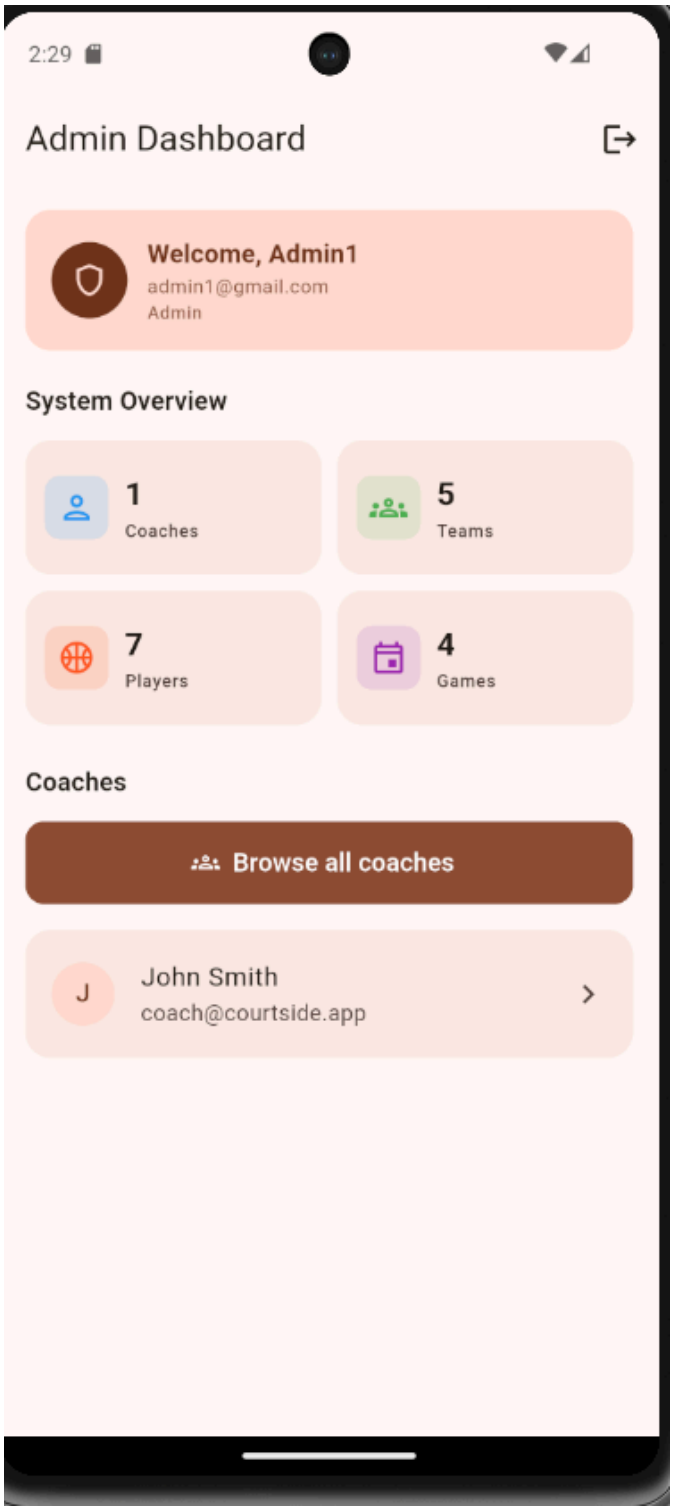

2:27
Ghost Riders
Ghost Riders
Fire Beach
Season
2026 Spring
Games
View, create, and record live games
Edit Player
Update this player's details.
Full name
Professor
Jersey number
22
Position
PG — Point Guard
Save Changes
Cancel

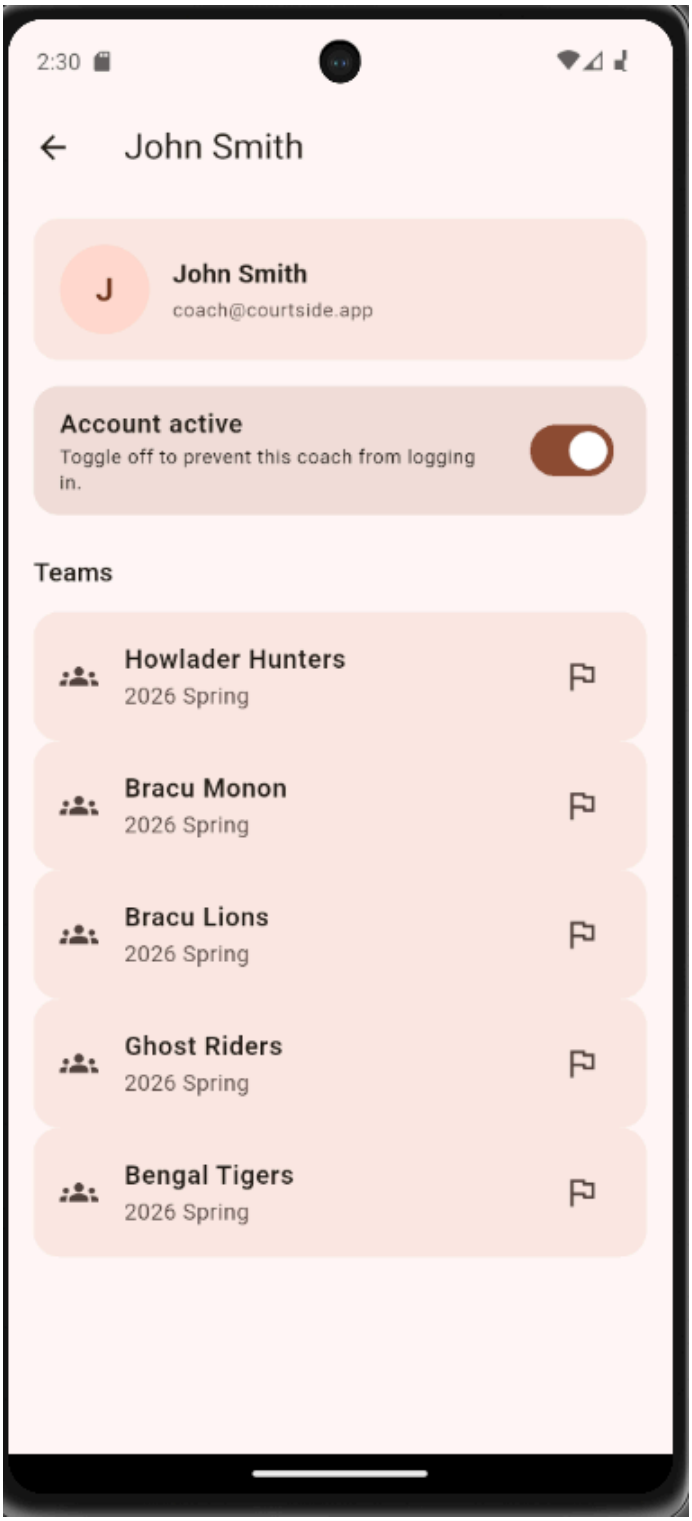

2:28
Ghost Riders
Ghost Riders
Fire Beach
Season
2026 Spring
Games
View, create, and record live games
Add Player
Enter the player's name, jersey number, and position.
Full name
Jersey number
Position
PG — Point Guard
Add Player
Cancel


2:28
Ghost Riders
Ghost Riders
Fire Beach
Season
2026 Spring
Games
View, create, and record live games
Edit Team
Update the team's details.
Team name
Ghost Riders
Season
2026 Spring
Home court
Fire Beach
Save Changes
Cancel

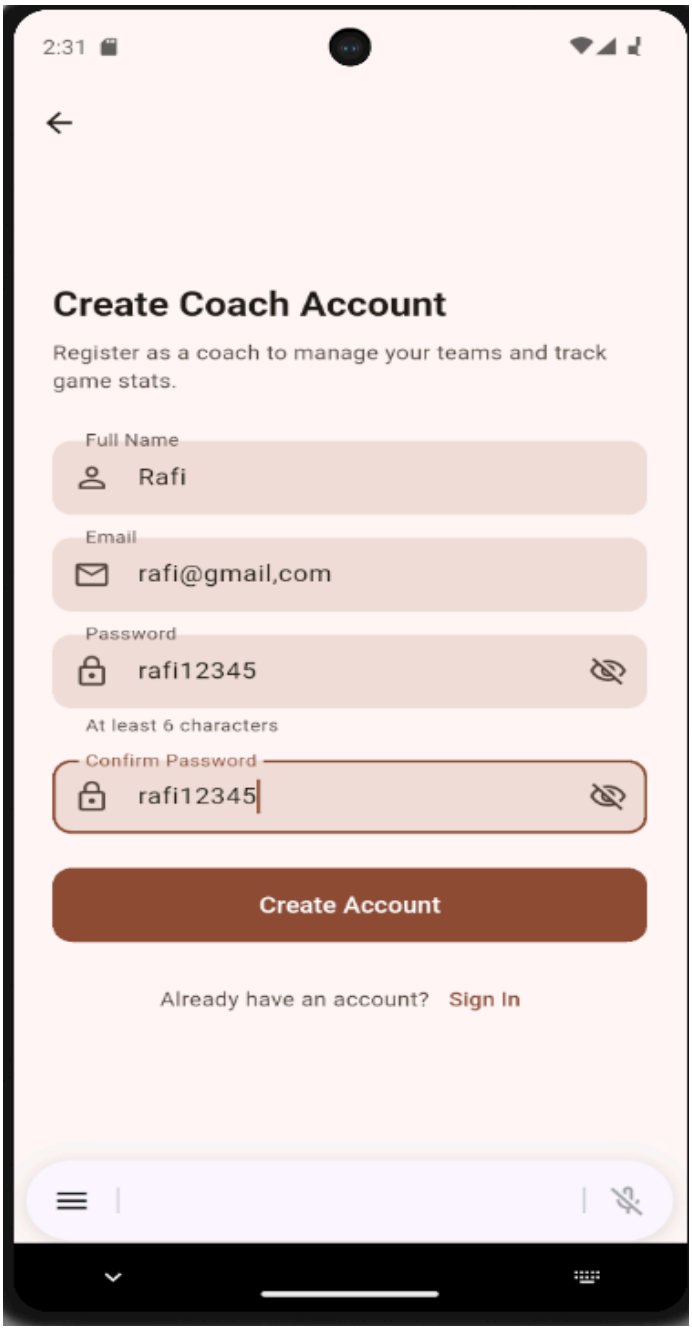

2:28
Ghost Riders
Ghost Riders
Fire Beach
Season
2026 Spring
Delete this team?
This will permanently delete the team along with all its players, games, and stats. This cannot be undone.
Cancel
Delete
23
LeBron James
Small Forward
24
sany
Point Guard
Add Player

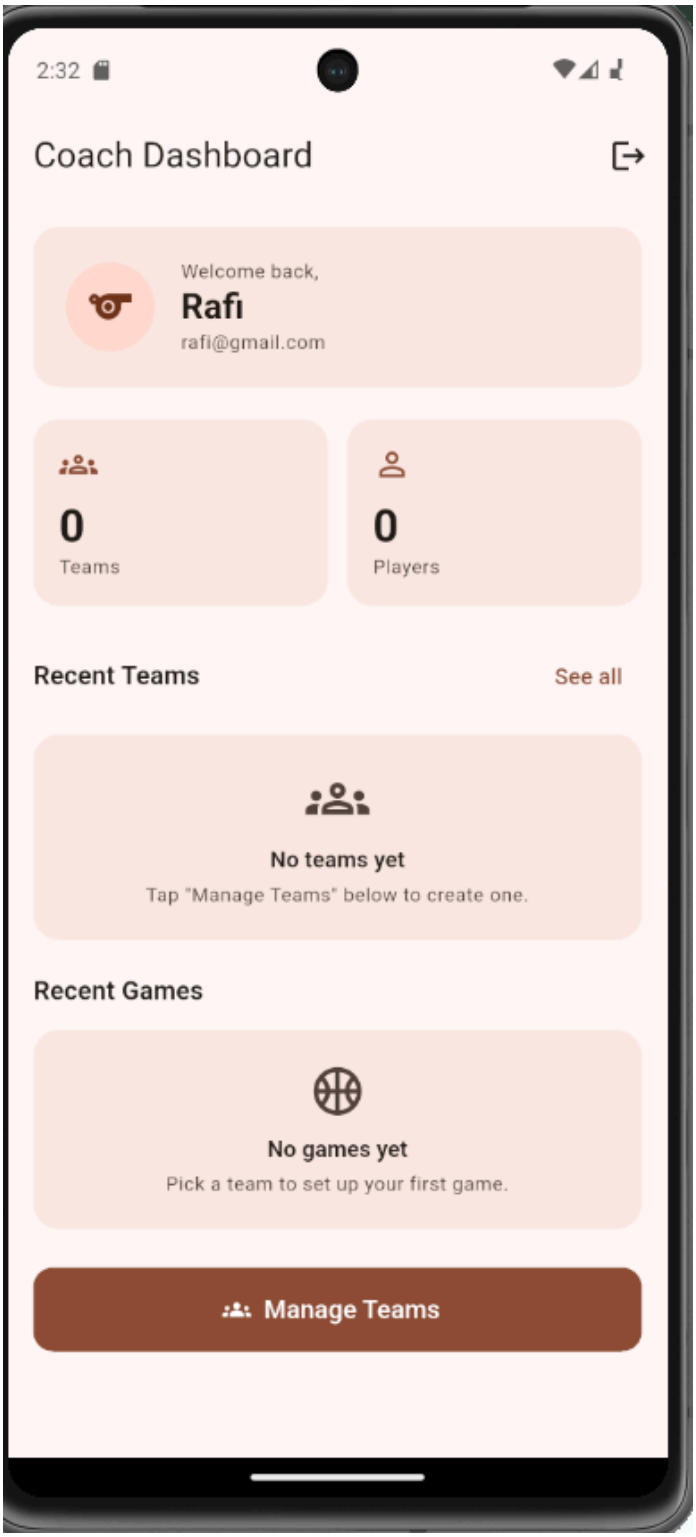
Admin Dash Board:




2:30
John Smith
J
John Smith
coach@courtside.app
Account active
Toggle off to prevent this coach from logging in.
Teams
Howlader Hunters
2026 Spring
Bracu Monon
2026 Spring
Bracu Lions
2026 Spring
Ghost Riders
2026 Spring
Bengal Tigers
2026 Spring

Create Coach Account:





<u>Online Resources used</u>:

a) **Reference:**

- Flutter documentation — https://docs.flutter.dev
- drift ORM documentation — https://drift.simonbinder.eu
- Riverpod documentation — https://riverpod.dev
- go_router documentation — https://pub.dev/packages/go_router
- fl_chart examples and API reference — https://github.com/imaNNeo/fl_chart
- pdf and printing package documentation — https://pub.dev/packages/pdf
- W3Schools SQL reference for query design

b) **Stackoverflow / GitHub:**

- Stack Overflow threads on drift streaming queries and Riverpod state management
- GitHub issues on share_plus 10.x API and Printing package's share vs print intents
- Original freelance project posting on Upwork (basketball stat tracker brief) that inspired the concept

Future Enhancements:

Following enhancements can be added to the current system which will improve the system:

a) **Real-time substitutions tracking** — currently the live game screen treats all roster players as on-court at all times. Adding bench/active toggling would allow more accurate plus-minus stats.

b) **Foul tracking and free throws context** — record fouls per player and link them to free throw attempts. This would enable foul-out warnings during live games.

c) **Quarter / period-by-period stats** — segment game data by quarter so coaches can analyze performance trends within a single game.

d) **Cloud sync via Firebase** — currently all data is local SQLite. Syncing to Firebase would let coaches access their data across devices and let admins do remote oversight without sharing physical devices.

e) **Player photos** — allow each player profile to have an uploaded photo for visual recognition, especially helpful when a coach manages large rosters or multiple teams.

f) **Practice mode** — log practice attendance and individual drill stats separately from games. Currently only game data is tracked.

g) **Schedule/calendar view** — upcoming games on a monthly calendar with reminders, integrated with the existing game model.

h) **Export to CSV** — for coaches who want to do their own analysis in Excel or Google Sheets, alongside the existing PDF export.